

Let's go through **SQL Server Always On Availability Groups (AGs)** in a **real-time DBA-oriented manner**, step-by-step, covering:

1.  **Prerequisites**
2.  **Real-Time Use Cases**
3.  **Advantages**
4.  **Disadvantages / Limitations**
5.  **Common Issues and Resolutions**

1. Prerequisites of Always On Availability Groups

Before configuring AlwaysOn AG, several **infrastructure**, **SQL Server**, and **Windows** requirements must be met.

A. Infrastructure Prerequisites

Component	Requirement
Windows Server Edition	Windows Server 2012 or higher (Standard supports 1 DB per AG, Enterprise supports multiple DBs).
Domain	All replicas must be in the same Active Directory domain .
Failover Clustering Feature	Windows Server Failover Cluster (WSFC) must be installed and configured on all replicas.
Shared Storage	Not required (unlike Failover Cluster Instance). Each replica maintains its own copy of data .
Network Configuration	- Low-latency, high-bandwidth network between replicas.- Static IP for cluster name and listener.
DNS Access	Proper name resolution for replicas and listener.
Firewall Rules	Open ports (default: 1433 for SQL, and dynamic ports for endpoints).

B. SQL Server-Level Prerequisites

Area	Details
Edition	- Enterprise Edition: Supports automatic failover, readable secondaries, multiple DBs per AG.- Standard Edition: Supports Basic AG (1 DB per group, 2 replicas max).
SQL Server Version	SQL Server 2012 and later (features mature in 2016, 2017, 2019+).
SQL Service Accounts	- Must be domain accounts . - Must have connect permissions for endpoints.- Preferably dedicated per instance .
Database State	- Databases must be in FULL Recovery Model . - At least one full backup taken before adding to AG.
Endpoints	Database Mirroring endpoints (TCP) created and secured with certificates or domain auth.
Quorum Configuration	Proper quorum model (Node Majority, Node + File Share Majority, etc.).

C. Optional but Recommended

- Synchronous replicas in **same data center**, asynchronous for **remote DR site**.
- Use **Dedicated NICs** for data replication traffic.
- Configure **listener** for client transparent redirection.
- Enable **Always On Feature** in SQL Server Configuration Manager.

2. Real-Time Use Cases

Here's how DBAs and organizations use AlwaysOn in the real world:

Use Case	Description	Real-Time Example
High Availability (HA)	Protects against local instance failure.	Primary in Data Center A, Secondary in same DC with synchronous commit and automatic failover.
Disaster Recovery (DR)	Ensures business continuity if entire site goes down.	DR replica in another data center or Azure VM with asynchronous commit.

Use Case	Description	Real-Time Example
Read Scale-Out	Offload read-only queries to secondary replicas.	Reporting or Power BI reads from secondary replica.
Backup Offloading	Backups can run on secondary to reduce primary load.	Nightly full backups taken from secondary replica.
Zero Data Loss Failover	Synchronous mode ensures no data loss during failover.	Finance/Banking systems.
Maintenance Flexibility	Patch one replica while others serve production traffic.	Apply OS/SQL patches on secondary first → failover → patch primary.

3. Advantages of Always On AG

Benefit	Description
No Shared Storage Dependency	Each node keeps its own copy — no single point of failure.
Fast Automatic Failover	Sub-second detection and failover (in sync mode).
Multiple Secondaries (up to 8)	Enhances HA and DR coverage.
Readable Secondaries	Improves performance for reporting and backups.
Granular DB-Level Protection	Not instance-level (unlike FCI).
Supports Hybrid Cloud	Combine on-premises and Azure (Azure IaaS or Managed Instance).
Transparent Client Redirection	Application auto-connects to new primary via listener.
Improved RTO/RPO	Better recovery time and point objectives.

4. Disadvantages / Limitations

Limitation	Description / Impact
Enterprise Licensing Cost	Full AGs require Enterprise Edition (costly).
No Cross-Database Transactions Support	Fails over per-database, not per-instance.
DBs in AG Must Share Same Log Chain	All DBs must have valid log sequence.
Manual Failover for Asynchronous Replicas	DR site failover is manual.
High Network Bandwidth Requirement	Especially in synchronous mode for large workloads.
No AG on Simple Recovery DBs	FULL recovery mandatory.
Maintenance Overhead	Complex to manage, especially with multiple replicas and listeners.
TempDB Not Replicated	Local to each replica; behavior can differ.
Cluster Dependency	WSFC issues can cause AG instability.

5. Common Issues and Resolutions

Issue	Root Cause	Resolution
Database not synchronizing	Log backup chain broken or endpoint connectivity issue.	Reinitialize replica: backup log from primary → restore with NORECOVERY → rejoin AG.
Automatic failover not happening	Replica not in synchronous mode or not set for automatic failover.	Verify both replicas are synchronous and configured for auto failover.

Issue	Root Cause	Resolution
Listener not accessible	DNS or firewall issue.	Check listener DNS record, port 1433 open, and correct IP binding.
Secondary lag high	Network latency or insufficient I/O on secondary.	Switch to async mode for DR, optimize network/I/O, or upgrade NICs.
Cluster goes to resolving state	Quorum lost or WSFC misconfigured.	Review quorum configuration; use witness (file share/cloud witness).
Data loss after failover (async mode)	Expected behavior (async replication).	For zero data loss, use synchronous commit.
Login/Job Mismatch	Logins, SQL Agent jobs, and linked servers are not auto-synced.	Use scripts or automation (DBATools) to sync logins/jobs across replicas.
Port Conflicts on Endpoints	Same port used by multiple instances.	Assign unique ports for each replica endpoint.
Replica failed to connect to endpoint	SQL Service account permission issue.	Grant CONNECT permission on endpoint to service accounts.

Bonus: Monitoring & Maintenance Tips

- Regularly check:
 - sys.dm_hadr_availability_replica_states
 - sys.dm_hadr_database_replica_states
 - SQL Agent jobs for AG failover notifications
- Use **Extended Events** for AG state change tracking.
- Automate logins/jobs sync via PowerShell or DBATools (Copy-DbaLogin, Copy-DbaAgentJob).
- Always take **regular full & log backups** even with AGs.
- Patch order: **secondary first → failover → patch primary → failback.**

Summary

Category	Key Takeaway
Purpose	High Availability, Disaster Recovery, Read Scale-out
Main Requirement	WSFC + Full Recovery Model
Best Use Case	Enterprise-grade mission-critical systems
Best Practices	Use synchronous for HA, async for DR, automate monitoring
Common Mistakes	Forgetting FULL recovery, wrong quorum, not syncing logins/jobs